

EXIT EXAM TUTORIAL

SPECIFIC OBJECTIVES (Exam blueprint):

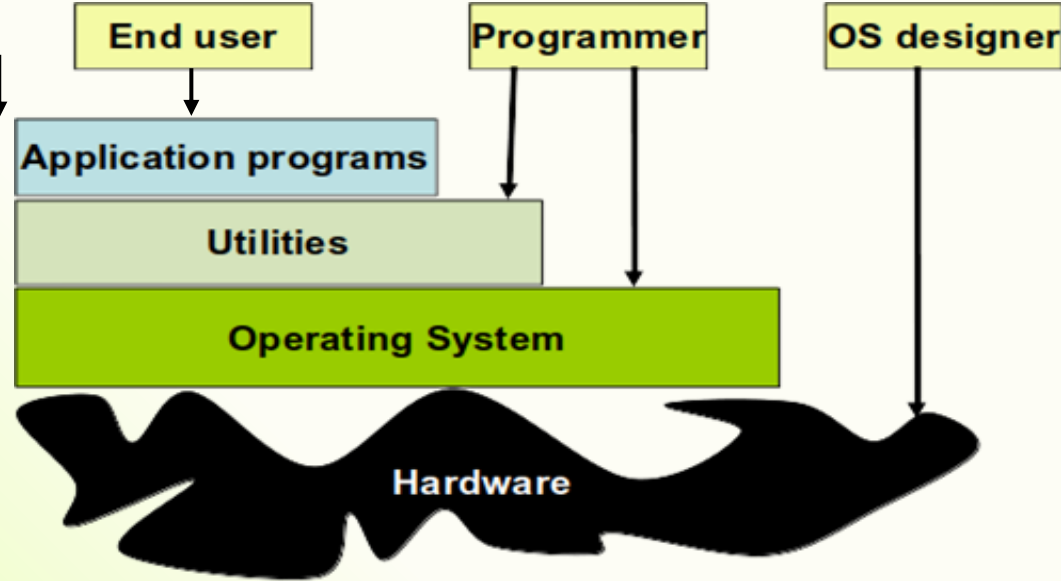
- **Recall the structure and functionality of modern Operating Systems**
- **Explain conditions that lead to deadlock and memory management, I/O, mass storage, and file management issues**
- **Describe security and protection measures used in operating systems**
- **Use knowledge of process management and synchronization for software development**
- **Compare the various algorithms on which the core functions of the Operating Systems are built**



What is operating system?

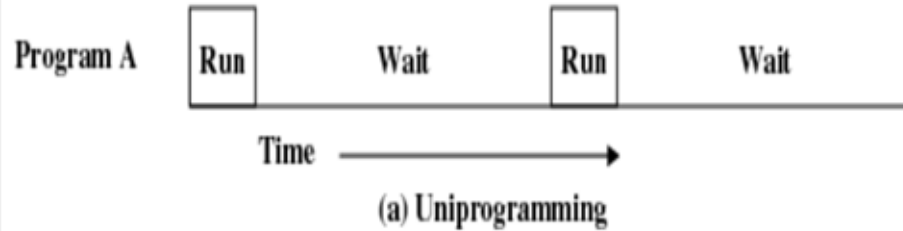
Recall the structure and functionality of modern Operating Systems

- A program that
 - controls the execution of a program and
 - acts as an intermediary between a user and the hardware



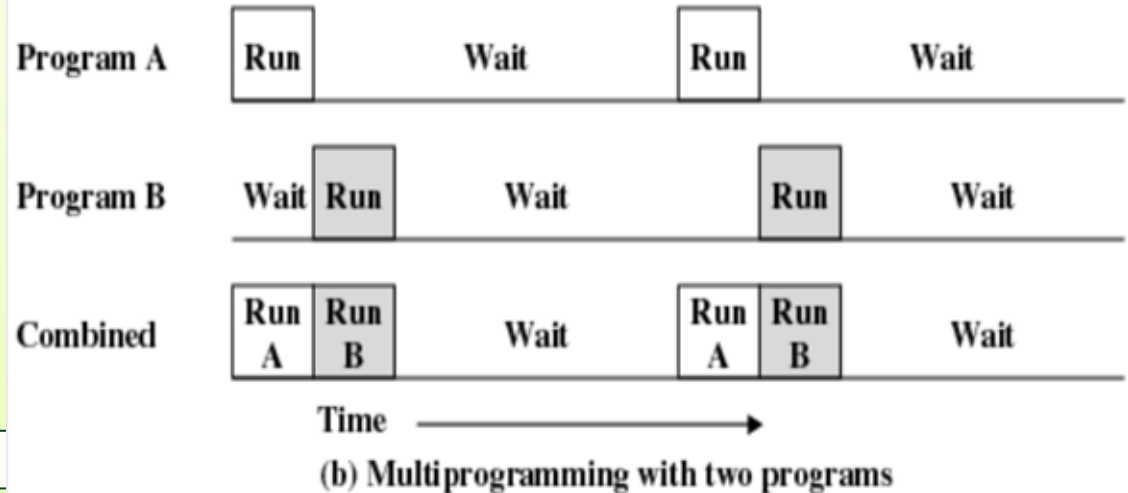
Uniprogramming vs Multiprogramming

Recall the structure and
functionality of modern
Operating Systems



- Processor can't switch to another process; it must wait for I/O instruction to complete before proceeding

When one job needs to wait for I/O, the processor can switch to the other job

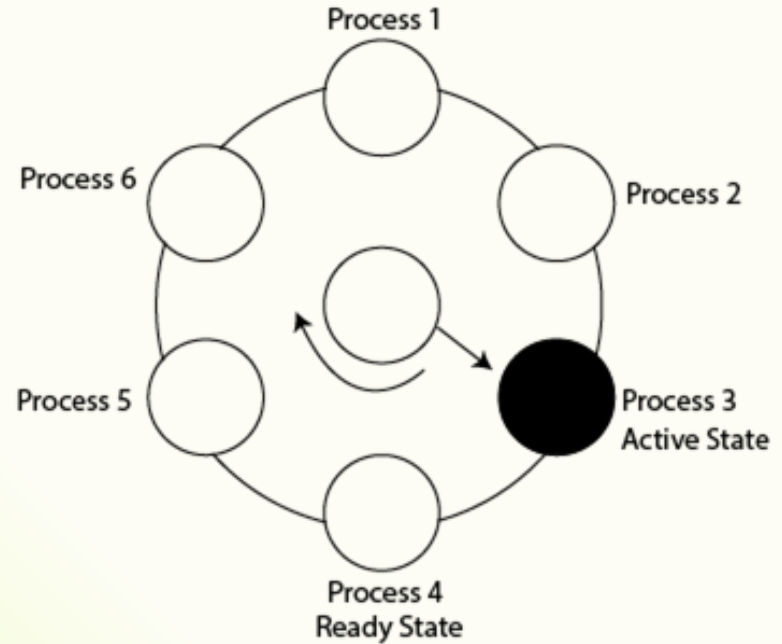


Timesharing

OS decides to stop running process and start running another

When to stop a process:
Time quantum

Recall the structure and functionality of modern Operating Systems



Services of OS

Recall the structure and
functionality of modern
Operating Systems

Make the computer convenient to users:

Program development

Program execution

Access to I/O

Controlled access to files

System access

Error detection and response

Accounting



Services of OS

Recall the structure and
functionality of modern
Operating Systems

Resources management:

Process management

create/terminate process

Process synchronization

Process scheduling

Interprocess scheduling



Services of OS

Recall the structure and
functionality of modern
Operating Systems

Resources management:

Memory management

Allocation and deallocation of memory

Virtual memory management

Paging

Protection

Issues in memory management

Fragmentation: internal vs external

Memory leaks and trashing



Services of OS

Recall the structure and
functionality of modern
Operating Systems

Resources management:

File management

- Create/delete files

- Organize files into directories/indexes

- Manage secondary storage

 - Disk allocation algorithms

 - Disk scheduling algorithms: FCFS, SCAN, ...

 - Disk caching

- File protection

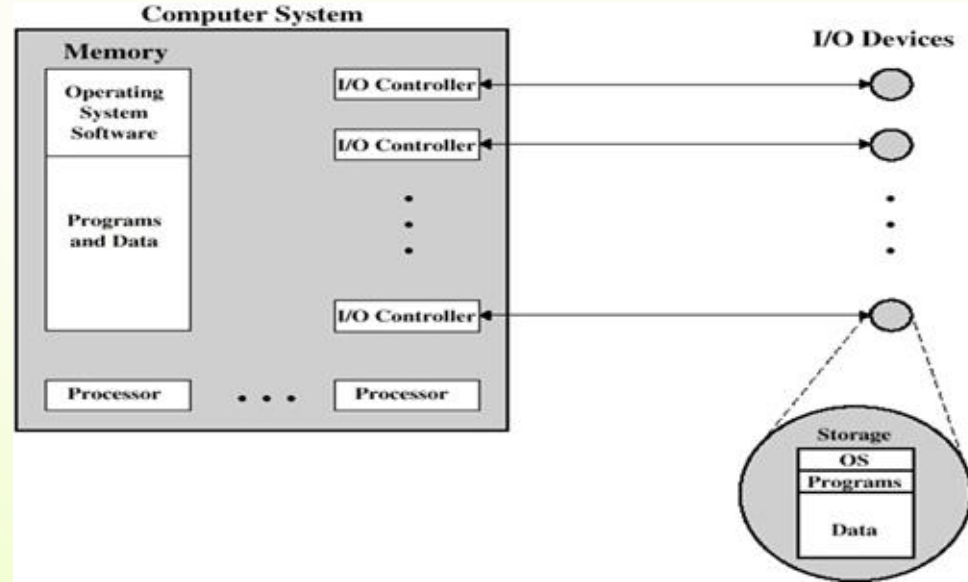
- File types: FAT, NTFS, EXT4



Services of OS

Recall the structure and functionality of modern Operating Systems

Resources management:
I/O or device management
Controlling devices
Data transfer
Device driver
Interrupt handling
Device varieties:
Speed
Command
Buffering
DMA

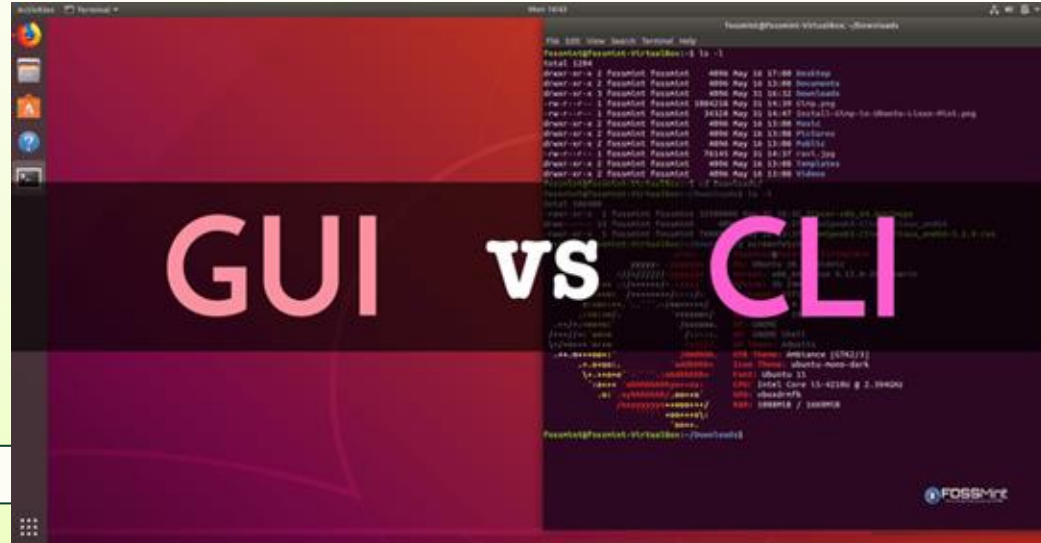


Services of OS

Recall the structure and functionality of modern Operating Systems

Resources management:

- Security and protection
 - Authentication
 - Authorization
 - Data security
- User interface
 - CLI
 - GUI



Deadlock

Deadlock and System Resource Management Issues

Mutual Exclusion

Resources cannot be shared.

Hold and Wait

A process holds one resource while waiting for another.

No Preemption

Resources cannot be forcibly removed.

Circular Wait

Processes form a circular chain waiting for resources.



Security

Authentication

Password

Biometrics

Smart cards

Authorization

RWX $\Rightarrow$ Owner, Group, Others

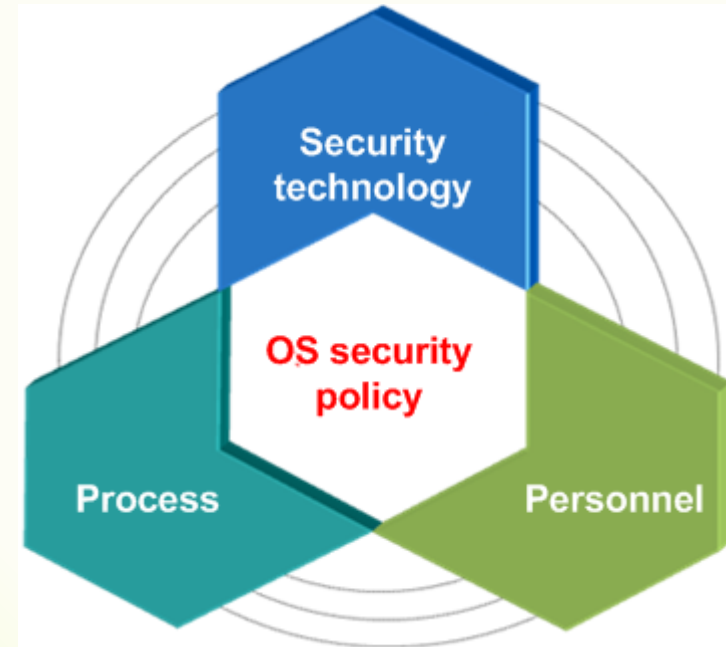
Protection

ACL

Encryption

Antivirus

Describe security and protection measures used in operating systems



Process

Use knowledge of process management
and synchronization for software
development

It is a program in execution, has

Process ID, PPID, UID

Context

Memory space (address space)

States

New, Ready, Running, Waiting, Terminated

Two blocks:

Process control block - used by OS

Process image - contains program, data, heap



Process Synchronization

Use knowledge of process management and synchronization for software development

Control problems:

Deadlock

Starvation

Mutual exclusion

Algorithms: lock variable, semaphores, sleep & wakeup, monitors, etc.

Classical problems:

Producer & consumer

Sleeping barber

Dinning philosophes

Readers and writers



Algorithms

Compare the various algorithms on which the core functions of the Operating Systems are built

CPU scheduling

FCFS, SJF, SRTN, RR, Priority

Memory allocation

Dynamic partitioning: First-fit, Next-fit, Best-fit, Worst-fit,
Segmentation

Paging

Disc scheduling algorithms

FCFS, SSTF, C-Scan, LOOK,

Page replacement

Optimal, FIFO, NRU, LRU, Clock



Good luck!

